

Web Appendices for “Covariate-adjusted win statistics in randomized clinical trials with ordinal outcomes”

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This documentation presented the web appendices and tables not shown in the manuscript.

Web Appendix A: Consistent estimators for variance of unadjusted estimator

A1. Estimation of variance for win probability and loss probability

For unadjusted estimator $\hat{\tau}_1^{\text{UNADJ}}$, we have

$$\hat{\tau}_1^{\text{UNADJ}} = \frac{1}{n_1 n_0} \sum_{i=1}^{n_1} \sum_{j=1}^{n_0} \omega_{ij} \mathbb{I}(Y_i > Y_j) = \frac{\sum_{i=1}^n \sum_{j=1}^n Z_i (1 - Z_j) \mathbb{I}(Y_i > Y_j)}{\sum_{i=1}^n \sum_{j=1}^n Z_i (1 - Z_j)},$$

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where $\omega_{ij} = \frac{Z_i(1-Z_j)}{\frac{1}{n_1} \sum_{i=1}^{n_1} Z_i \times \frac{1}{n_0} \sum_{j=1}^{n_0} (1-Z_j)}$. In main text, we have derived its asymptotic variance as $\sigma_1^2 = \frac{n}{n_1} \xi_{10} + \frac{n}{n_0} \xi_{01}$, $n = n_1 + n_0$ and

$$\begin{aligned}\xi_{10} &= \text{cov}(\omega_{ij}\mathbb{I}(Y_i > Y_j), \omega_{ik}\mathbb{I}(Y_i > Y_k)) \\ &= E[\omega_{ij}\mathbb{I}(Y_i > Y_j) \times \omega_{ik}\mathbb{I}(Y_i > Y_k)] - E[\omega_{ij}\mathbb{I}(Y_i > Y_j)]E[\omega_{ik}\mathbb{I}(Y_i > Y_k)] \\ &= P(Y_i(1) > Y_j(0) \& Y_i(1) > Y_k(0)) - [P(Y_i(1) > Y_j(0))]^2, \\ \xi_{01} &= \text{cov}(\omega_{ij}\mathbb{I}(Y_i > Y_j), \omega_{kj}\mathbb{I}(Y_k > Y_j)), \\ &= E[\omega_{ij}\mathbb{I}(Y_i > Y_j) \times \omega_{kj}\mathbb{I}(Y_k > Y_j)] - E[\omega_{ij}\mathbb{I}(Y_i > Y_j)]E[\omega_{kj}\mathbb{I}(Y_k > Y_j)] \\ &= P(Y_i(1) > Y_j(0) \& Y_k(1) > Y_j(0)) - [P(Y_i(1) > Y_j(0))]^2, \quad i, j, k = 1, \dots, n.\end{aligned}$$

To obtain consistent estimators for σ_1^2 , similar to the approach of Bebu and Lachin (2016), we denote $S_{i\cdot}^1 = \sum_{j=1}^{n_0} \omega_{ij}\mathbb{I}(Y_i > Y_j)$, $S_{\cdot j}^2 = \sum_{i=1}^{n_0} \omega_{ij}\mathbb{I}(Y_i < Y_j)$, $S_{i\cdot}^1 = \sum_{i=1}^{n_1} \omega_{ij}\mathbb{I}(Y_i > Y_j)$ and $S_{\cdot j}^2 = \sum_{i=1}^{n_1} \omega_{ij}\mathbb{I}(Y_i < Y_j)$. Then the above probabilities can be estimated as follows:

$$\begin{aligned}\hat{P}(Y_i(1) > Y_j(0)) &= \hat{E}[\mathbb{I}(Y_i(1) > Y_j(0))] = \hat{\tau}_1^{\text{UNADJ}} = \frac{\sum_{i=1}^n \sum_{j=1}^n Z_i(1-Z_j)\mathbb{I}(Y_i > Y_j)}{\sum_{i=1}^n \sum_{j=1}^n Z_i(1-Z_j)}, \\ \hat{P}(Y_i(1) > Y_j(0) \& Y_i(1) > Y_k(0)) &= \frac{\sum_{i=1}^{n_1} S_{i\cdot}^1(S_{i\cdot}^1 - 1)}{n_1 n_0 (n_0 - 1)} \\ &= \frac{\sum_{i=1}^{n_1} \left\{ \sum_{j=1}^{n_0} \omega_{ij}\mathbb{I}(Y_i > Y_j) [\sum_{j=1}^{n_0} \omega_{ij}\mathbb{I}(Y_i > Y_j) - 1] \right\}}{n_1 n_0 (n_0 - 1)}, \\ &= \frac{\sum_{i=1}^{n_1} \left[\sum_{j=1}^{n_0} \omega_{ij}\mathbb{I}(Y_i > Y_j) \right]^2}{n_1 n_0 (n_0 - 1)} - \frac{1}{n_0 - 1} \hat{P}(Y_i(1) > Y_j(0)), \\ \hat{P}(Y_i(1) > Y_j(0) \& Y_k(1) > Y_j(0)) &= \frac{\sum_{j=1}^{n_0} S_{\cdot j}^1(S_{\cdot j}^1 - 1)}{n_0 n_1 (n_1 - 1)} \\ &= \frac{\sum_{j=1}^{n_0} \left\{ \sum_{i=1}^{n_1} \omega_{ij}\mathbb{I}(Y_i > Y_j) [\sum_{i=1}^{n_1} \omega_{ij}\mathbb{I}(Y_i > Y_j) - 1] \right\}}{n_0 n_1 (n_1 - 1)} \\ &= \frac{\sum_{j=1}^{n_0} \left[\sum_{i=1}^{n_1} \omega_{ij}\mathbb{I}(Y_i > Y_j) \right]^2}{n_0 n_1 (n_1 - 1)} - \frac{1}{n_1 - 1} \hat{P}(Y_i(1) > Y_j(0)).\end{aligned}$$

Therefore, a consistent variance estimator for $\hat{\tau}_1^{\text{UNADJ}}$ is

$$\begin{aligned}\widehat{\text{var}}(\hat{\tau}_1^{\text{UNADJ}}) &= \frac{1}{n} \left\{ \frac{n}{n_1} \hat{P}(Y_i(1) > Y_j(0) \& Y_i(1) > Y_k(0)) - \frac{n}{n_1} [\hat{P}(Y_i(1) > Y_j(0))]^2 \right. \\ &\quad \left. + \frac{n}{n_0} \hat{P}(Y_i(1) > Y_j(0) \& Y_k(1) > Y_j(0)) - \frac{n}{n_0} [\hat{P}(Y_i(1) > Y_j(0))]^2 \right\} \\ &= \frac{1}{n_1} \hat{P}(Y_i(1) > Y_j(0) \& Y_i(1) > Y_k(0)) + \frac{1}{n_0} \hat{P}(Y_i(1) > Y_j(0) \& Y_k(1) > Y_j(0)) - \frac{n}{n_1 n_0} [\hat{P}(Y_i(1) > Y_j(0))]^2.\end{aligned}$$

By replacing operator “>” by “<” in the above estimators, i.e., replacing “>” in $\hat{P}(Y_i(1) > Y_j(0))$, $\hat{P}(Y_i(1) > Y_j(0) \& Y_i(1) > Y_k(0))$ and $\hat{P}(Y_i(1) > Y_j(0) \& Y_k(1) > Y_j(0))$ by “<”, we can obtain $\widehat{\text{var}}(\hat{\tau}_{-1}^{\text{UNADJ}})$, where $\hat{\tau}_{-1}^{\text{UNADJ}} = \frac{\sum_{i=1}^n \sum_{j=1}^n Z_i(1-Z_j)\mathbb{I}(Y_i < Y_j)}{\sum_{i=1}^n \sum_{j=1}^n Z_i(1-Z_j)}$.

A2. Estimation of covariance between win probability and loss probability

For the asymptotic covariance $\sigma_{12} = \frac{n}{n_1} \xi_{10}^{12} + \frac{n}{n_0} \xi_{01}^{12}$, where

$$\begin{aligned}
\xi_{10}^{12} &= \text{cov}(\omega_{ij}\mathbb{I}(Y_i > Y_j), \omega_{ik}\mathbb{I}(Y_i < Y_k)), \\
&= E[\omega_{ij}\mathbb{I}(Y_i > Y_j) \times \omega_{ik}\mathbb{I}(Y_i < Y_k)] - E[\omega_{ij}\mathbb{I}(Y_i > Y_j)]E[\omega_{ik}\mathbb{I}(Y_i < Y_k)] \\
&= P(Y_i(1) > Y_j(0) \& Y_i(1) < Y_k(0)) - P(Y_i(1) > Y_j(0))P(Y_i(1) < Y_k(0)), \\
\xi_{01}^{12} &= \text{cov}(\omega_{ij}\mathbb{I}(Y_i > Y_j), \omega_{kj}\mathbb{I}(Y_k < Y_j)), \\
&= E[\omega_{ij}\mathbb{I}(Y_i > Y_j) \times \omega_{kj}\mathbb{I}(Y_k < Y_j)] - E[\omega_{ij}\mathbb{I}(Y_i > Y_j)]E[\omega_{kj}\mathbb{I}(Y_k < Y_j)] \\
&= P(Y_i(1) > Y_j(0) \& Y_k(1) < Y_j(0)) - P(Y_i(1) > Y_j(0))P(Y_k(1) < Y_j(0)).
\end{aligned}$$

Similarly, we can obtain their consistent estimators as

$$\begin{aligned}
\hat{P}(Y_i(1) < Y_j(0)) &= \frac{\sum_{i=1}^n \sum_{j=1}^n Z_i(1 - Z_j)\mathbb{I}(Y_i < Y_j)}{\sum_{i=1}^n \sum_{j=1}^n Z_i(1 - Z_j)}, \\
\hat{P}(Y_i(1) > Y_j(0) \& Y_i(1) < Y_k(0)) &= \frac{\sum_{i=1}^{n_1} S_{i\cdot}^1 S_{i\cdot}^2}{n_1 n_0 (n_0 - 1)} \\
&= \frac{\sum_{i=1}^{n_1} \left\{ \sum_{j=1}^{n_0} \omega_{ij}\mathbb{I}(Y_i > Y_j) [\sum_{j=1}^{n_0} \omega_{ij}\mathbb{I}(Y_i < Y_j)] \right\}}{n_1 n_0 (n_0 - 1)}, \\
\hat{P}(Y_i(1) > Y_j(0) \& Y_k(1) < Y_j(0)) &= \frac{\sum_{j=1}^{n_0} S_{\cdot j}^1 S_{\cdot j}^2}{n_0 n_1 (n_1 - 1)} \\
&= \frac{\sum_{j=1}^{n_0} \left\{ \sum_{i=1}^{n_1} \omega_{ij}\mathbb{I}(Y_i > Y_j) [\sum_{i=1}^{n_1} \omega_{ij}\mathbb{I}(Y_i < Y_j)] \right\}}{n_0 n_1 (n_1 - 1)}.
\end{aligned}$$

Therefore, the covariance estimator between $\hat{\tau}_1^{\text{UNADJ}}$ and $\hat{\tau}_{-1}^{\text{UNADJ}}$ is $\frac{1}{n} \hat{\sigma}_{12}$, which can be computed as

$$\begin{aligned}
\text{cov}(\hat{\tau}_1^{\text{UNADJ}}, \hat{\tau}_{-1}^{\text{UNADJ}}) &= \frac{1}{n} \left\{ \frac{n}{n_1} \hat{P}(Y_i(1) > Y_j(0) \& Y_i(1) < Y_k(0)) - \frac{n}{n_1} \hat{P}(Y_i(1) > Y_j(0)) \hat{P}(Y_i(1) < Y_j(0)) \right. \\
&\quad \left. + \frac{n}{n_0} \hat{P}(Y_i(1) > Y_j(0) \& Y_k(1) < Y_j(0)) - \frac{n}{n_0} \hat{P}(Y_i(1) > Y_j(0)) \hat{P}(Y_k(1) < Y_j(0)) \right\} \\
&= \frac{1}{n_1} \hat{P}(Y_i(1) > Y_j(0) \& Y_i(1) < Y_k(0)) + \frac{1}{n_0} \hat{P}(Y_i(1) > Y_j(0) \& Y_k(1) < Y_j(0)) \\
&\quad - \frac{n}{n_1 n_0} \hat{P}(Y_i(1) > Y_j(0)) \hat{P}(Y_i(1) < Y_j(0)).
\end{aligned}$$

A3. Bias-reduced variance estimator suggested by Ozenne et al.(2021)

It is noted that according to the Bebu and Lachin (2016), we constrain the comparisons in the variance estimation to strictly different individuals, i.e., $P[Y_i(1) > Y_j(0) \& Y_i(1) > Y_k(0)]$, which will be consistent with true variance under large sample size. However, Ozenne et al.(2021) showed that this leads to bias in small samples and to potentially negative variance values for $\hat{\tau}_1^{\text{UNADJ}}$. And they suggested a variance estimator based on first and second order H-decomposition of U-statistics that allows comparisons in the variance estimation to also come from the same individual, which has less bias and can only be positive.

Actually, their first-order H-decomposition estimator of variance of $\hat{\tau}_1^{\text{UNADJ}}$ is closely related to the estimator proposed by Bebu and Lachin (2016). To make the variance estimation of $\hat{\tau}_1^{\text{UNADJ}}$ perform well under small sample size, we can update the variances of $\hat{\tau}_1^{\text{UNADJ}}$, $\hat{\tau}_{-1}^{\text{UNADJ}}$ and their covariance estimation following the variance suggested by Ozenne et al.(2021), named as bias-reduced variance estimator in this article. To obtain the bias-reduced variance estimator, according to the formula (4) in Ozenne et al.(2021), we need to modify the expressions of $\hat{P}(Y_i(1) > Y_j(0) \& Y_i(1) > Y_k(0))$, $\hat{P}(Y_i(1) > Y_j(0) \& Y_k(1) > Y_j(0))$, $\hat{P}(Y_i(1) > Y_j(0) \& Y_i(1) < Y_k(0))$ and $\hat{P}(Y_i(1) > Y_j(0) \& Y_k(1) < Y_j(0))$ in Web Appendix A1 and A2 as follows, that is,

$$\begin{aligned}
\hat{P}(Y_i(1) > Y_j(0) \& Y_i(1) > Y_k(0)) &= \frac{\sum_{i=1}^{n_1} S_{i\cdot}^1(S_{i\cdot}^1)}{n_1 n_0^2} = \frac{\sum_{i=1}^{n_1} \left\{ \sum_{j=1}^{n_0} \omega_{ij} \mathbb{I}(Y_i > Y_j) [\sum_{j=1}^{n_0} \omega_{ij} \mathbb{I}(Y_i > Y_j)] \right\}}{n_1 n_0^2}, \\
&= \frac{\sum_{i=1}^{n_1} \left[\sum_{j=1}^{n_0} \omega_{ij} \mathbb{I}(Y_i > Y_j) \right]^2}{n_1 n_0^2}, \\
\hat{P}(Y_i(1) > Y_j(0) \& Y_k(1) > Y_j(0)) &= \frac{\sum_{j=1}^{n_0} S_{\cdot j}^1(S_{\cdot j}^1)}{n_0 n_1^2} = \frac{\sum_{j=1}^{n_0} \left\{ \sum_{i=1}^{n_1} \omega_{ij} \mathbb{I}(Y_i > Y_j) [\sum_{i=1}^{n_1} \omega_{ij} \mathbb{I}(Y_i > Y_j)] \right\}}{n_0 n_1^2}, \\
&= \frac{\sum_{j=1}^{n_0} \left[\sum_{i=1}^{n_1} \omega_{ij} \mathbb{I}(Y_i > Y_j) \right]^2}{n_0 n_1^2}, \\
\hat{P}(Y_i(1) > Y_j(0) \& Y_i(1) < Y_k(0)) &= \frac{\sum_{i=1}^{n_1} S_{i\cdot}^1 S_{i\cdot}^2}{n_1 n_0^2} \\
&= \frac{\sum_{i=1}^{n_1} \left\{ \sum_{j=1}^{n_0} \omega_{ij} \mathbb{I}(Y_i > Y_j) [\sum_{j=1}^{n_0} \omega_{ij} \mathbb{I}(Y_i < Y_j)] \right\}}{n_1 n_0^2}, \\
\hat{P}(Y_i(1) > Y_j(0) \& Y_k(1) < Y_j(0)) &= \frac{\sum_{j=1}^{n_0} S_{\cdot j}^1 S_{\cdot j}^2}{n_0 n_1^2} \\
&= \frac{\sum_{j=1}^{n_0} \left\{ \sum_{i=1}^{n_1} \omega_{ij} \mathbb{I}(Y_i > Y_j) [\sum_{i=1}^{n_1} \omega_{ij} \mathbb{I}(Y_i < Y_j)] \right\}}{n_0 n_1^2}.
\end{aligned}$$

Therefore, the bias-reduced variance and covariance estimators suggested by Ozenne et al.(2021) can be computed as

$$\begin{aligned}
\widehat{\text{var}}(\hat{\tau}_1^{\text{UNADJ}}) &= \frac{1}{n} \left\{ \frac{n}{n_1} \hat{P}(Y_i(1) > Y_j(0) \& Y_i(1) > Y_k(0)) - \frac{n}{n_1} [\hat{P}(Y_i(1) > Y_j(0))]^2 \right. \\
&\quad \left. + \frac{n}{n_0} \hat{P}(Y_i(1) > Y_j(0) \& Y_k(1) > Y_j(0)) - \frac{n}{n_0} [\hat{P}(Y_i(1) > Y_j(0))]^2 \right\}, \\
\widehat{\text{cov}}(\hat{\tau}_1^{\text{UNADJ}}, \hat{\tau}_{-1}^{\text{UNADJ}}) &= \frac{1}{n} \left\{ \frac{n}{n_1} \hat{P}(Y_i(1) > Y_j(0) \& Y_i(1) < Y_k(0)) - \frac{n}{n_1} \hat{P}(Y_i(1) > Y_j(0)) \hat{P}(Y_i(1) < Y_k(0)) \right. \\
&\quad \left. + \frac{n}{n_0} \hat{P}(Y_i(1) > Y_j(0) \& Y_k(1) < Y_j(0)) - \frac{n}{n_0} \hat{P}(Y_i(1) > Y_j(0)) \hat{P}(Y_i(1) < Y_j(0)) \right\},
\end{aligned}$$

where $\hat{P}(Y_i(1) > Y_j(0)) = \hat{E}[\mathbb{I}(Y_i(1) > Y_j(0))] = \hat{\tau}_1^{\text{UNADJ}} = \frac{\sum_{i=1}^n \sum_{j=1}^n Z_i(1-Z_j) \mathbb{I}(Y_i > Y_j)}{\sum_{i=1}^n \sum_{j=1}^n Z_i(1-Z_j)}$, $\hat{P}(Y_i(1) < Y_j(0)) = \hat{E}[\mathbb{I}(Y_i(1) < Y_j(0))] = \hat{\tau}_{-1}^{\text{UNADJ}} = \frac{\sum_{i=1}^n \sum_{j=1}^n Z_i(1-Z_j) \mathbb{I}(Y_i < Y_j)}{\sum_{i=1}^n \sum_{j=1}^n Z_i(1-Z_j)}$.

Web Appendix B: Proof for Theorem 1

B1. Proof for Theorem 1(a)

Note that the stabilized weighting estimator $\hat{\tau}_1^h$ defined in section 3.3 of main text, that is,

$$\hat{\tau}_1^h = \binom{n}{2}^{-1} \sum_{i=1}^n \sum_{j>i}^n q^h(\mathcal{O}_i, \mathcal{O}_j; \hat{\beta}),$$

is a U-statistic, where $q^h(\mathcal{O}_i, \mathcal{O}_j; \hat{\beta}) = \frac{1}{2} [Z_i(1 - Z_j)\mathbb{I}(Y_i > Y_j)\omega(X_i, X_j) + Z_j(1 - Z_i)\mathbb{I}(Y_i < Y_j)\omega(X_j, X_i)] / \left[\frac{1}{n(n-1)} \sum_{i=1}^n \sum_{j=1}^n Z_i(1 - Z_j)\omega(X_i, X_j) \right]$. For balanced weighting $\omega(\cdot, \cdot)$, $q^h(\mathcal{O}_i, \mathcal{O}_j; \beta)$ is a real-valued symmetric function. It is easy to see that $(E \times E)q^h(\cdot, \cdot; \beta)^2 < \infty$ and the function $q^h(\mathcal{O}_i, \mathcal{O}_j; \beta)$ is continuously differentiable with respect to β on Φ_δ , where $\Phi_\delta = \{\beta \in \Phi : \|\beta - \beta_*\| < \delta\}$ with β_* as the true value of β , and Φ is the range of β with $\|\cdot\|$ denotes the Euclidean norm and $\delta > 0$.

Therefore, by the Lemma S1 of Mao (2018), it is clear that the influence function of $\hat{\tau}_1^h$ is

$$\phi^h(\mathcal{O}) = 2\{q_1^h(\mathcal{O}; \beta) - \tau_1^h\} + \frac{\partial}{\partial \beta^\top}(\mathbb{E} \times \mathbb{E})q^h(\mathcal{O}, \mathcal{O}; \beta)\tilde{l}_\beta(\mathcal{O}; \beta), \quad (\text{A.1})$$

where $q_1^h(\mathcal{O}; \beta) = \mathbb{E}q^h(\cdot, \mathcal{O}; \beta) = \frac{1}{2} \left[\frac{Z}{e(X)}g_1(Y) + \frac{1-Z}{1-e(X)}g_0(Y) \right]$ with $g_1(Y) = \int \mathbb{I}(Y > s)\nu_0(ds)$ and $g_0(Y) = \int \mathbb{I}(Y < t)\nu_1(dt)$, and $\tilde{l}_\beta(\mathcal{O}; \beta)$ is the influence function for β . Since $\hat{\tau}_1^h$ is consistent with τ_1^h , the asymptotic variance of $n^{1/2}(\hat{\tau}_1^h - \tau_1^h)$ is $\text{var}(\phi^h(\mathcal{O}))$.

B2. Proof for Theorem 1(b)

To show $\phi^h(\mathcal{O}) = \chi^h(\mathcal{O}) - \prod\{\chi^h(\mathcal{O})|\Lambda_e\}$, where

$$\chi^h(\mathcal{O}) = \frac{Z}{e(X)}g_1(Y) + \frac{1-Z}{1-e(X)}g_0(Y) - 2\tau_1^h,$$

and $\prod\{\chi^h(\mathcal{O})|\Lambda_e\}$ is the projection of $\chi^h(\mathcal{O})$ onto Λ_e , the tangent space of $\mathcal{M}_e$. Note that according to Tsiatis (2006), we can write Λ_e and $\prod\{\chi^h(\mathcal{O})|\Lambda_e\}$ as

$$\Lambda_e = \{B^{1 \times p}S_\beta(\mathcal{O}; \beta) \mid \forall \text{ real matrices } B^{1 \times p}\}$$

and

$$\prod\{\chi^h(\mathcal{O})|\Lambda_e\} = \mathbb{E}(\chi^h(\mathcal{O})S_\beta^\top) \{\mathbb{E}(S_\beta S_\beta^\top)\}^{-1} S_\beta(\mathcal{O}; \beta),$$

respectively, where $S_\beta(\mathcal{O}; \beta)$ is the efficient score function of β . To see this, observe that $\phi^h(\mathcal{O}) = \chi^h(\mathcal{O}) - \prod\{\chi^h(\mathcal{O})|\Lambda_e\}$ is equivalent to

$$\frac{\partial}{\partial \beta^\top}(E \times E)q^h(\mathcal{O}, \mathcal{O}; \beta)\tilde{l}_\beta(\mathcal{O}; \beta) = -\prod\{\chi^h(\mathcal{O})|\Lambda_e\}. \quad (\text{A.2})$$

Note that the influence function $\tilde{l}_\beta(\mathcal{O}; \beta) = \{ES_\beta(\mathcal{O}; \beta)^{\otimes 2}\}^{-1} S_\beta(\mathcal{O}; \beta)$ and

$$\frac{\partial}{\partial \beta^\top}(E \times E)q^h(\mathcal{O}, \mathcal{O}; \beta)\tilde{l}_\beta(\mathcal{O}; \beta) = \tilde{l}_\beta^\top(\mathcal{O}; \beta) \frac{\partial}{\partial \beta}(E \times E)q^h(\mathcal{O}, \mathcal{O}; \beta).$$

So if we can show

$$\frac{\partial}{\partial \beta}(E \times E)q^h(\mathcal{O}, \mathcal{O}; \beta) = -2E \{q_1^h(\mathcal{O}; \beta) - \tau_1^h\} S_\beta(\mathcal{O}; \beta), \quad (\text{A.3})$$

(A.2) is straightforwardly proved since $2[q_1^h(\mathcal{O}; \beta) - \tau_1^h] = \chi^h(\mathcal{O})$.

To prove (A.3), write the density of $\mathcal{O}$ as $f(x; \beta)$ with respect to some dominating measure κ when the true parameter in $\mathcal{M}_e$ is β . Then for all β , we have

$$\int \int \{q^h(x_1, x_2; \beta) - \tau_1^h\} f(x_1; \beta) f(x_2; \beta) \kappa(dx_1) \kappa(dx_2) = 0.$$

Taking the derivative of both sides with respect to β , we obtain

$$\begin{aligned} 0 &= \frac{\partial}{\partial \beta} \int \int \{q^h(x_1, x_2; \beta) - \tau_1^h\} f(x_1; \beta) f(x_2; \beta) \kappa(dx_1) \kappa(dx_2) \\ &= \int \int \frac{\partial}{\partial \beta} [\{q^h(x_1, x_2; \beta) - \tau_1^h\} f(x_1; \beta) f(x_2; \beta) \kappa(dx_1) \kappa(dx_2)] \\ &= \frac{\partial}{\partial \beta} (E \times E)q^h(\cdot, \cdot; \beta) + \int \int \{q^h(x_1, x_2; \beta) - \tau_1^h\} \dot{f}(x_1; \beta) f(x_2; \beta) \kappa(dx_1) \kappa(dx_2) \\ &\quad + \int \int \{q^h(x_1, x_2; \beta) - \tau_1^h\} f(x_1; \beta) \dot{f}(x_2; \beta) \kappa(dx_1) \kappa(dx_2), \end{aligned} \quad (\text{A.4})$$

where $\dot{f}(x; \beta)$ is the derivative of $f(x; \beta)$ with respect to β . Note that $S_\beta(x; \beta) = \frac{\partial \log[f(x; \beta)]}{\partial \beta} = \frac{\dot{f}(x; \beta)}{f(x; \beta)}$, so the second term on the right-hand side of (A.4) is

$$\begin{aligned} &\int \int \{q^h(x_1, x_2; \beta) - \tau_1^h\} \dot{f}(x_1; \beta) f(x_2; \beta) \kappa(dx_1) \kappa(dx_2) \\ &= \int \int \{q^h(x_1, x_2; \beta) - \tau_1^h\} f(x_2; \beta) \kappa(dx_2) \dot{f}(x_1; \beta) \kappa(dx_1) \\ &= \int \{q_1^h(\mathcal{O}; \beta) - \tau_1^h\} \dot{f}(x_1; \beta) \kappa(dx_1) = E \{q_1^h(\mathcal{O}; \beta) - \tau_1^h\} S_\beta(\mathcal{O}; \beta). \end{aligned}$$

Similarly, we can also show that the third term of the right-hand side of (A.4) leads to the same identity. Therefore, rearranging the terms of (A.4), we obtain the result of (A.3) so that the influence function $\phi^h(\mathcal{O}) = \chi^h(\mathcal{O}) - \prod \{\chi^h(\mathcal{O}) | \Lambda_e\}$. Thus, by the multivariate Pythagorean theorem (Tsiatis, 2006), the variance of $\hat{\tau}_1^h$ is

$$\text{Var}(\hat{\tau}_1^h) = \text{Var}[\phi^h(\mathcal{O})] = \text{Var}[\chi^h(\mathcal{O}) - \prod \{\chi^h(\mathcal{O}) | \Lambda_e\}] = \text{Var}[\chi^h(\mathcal{O})] - \text{Var}[\prod \{\chi^h(\mathcal{O}) | \Lambda_e\}]. \quad (\text{A.5})$$

B3. Proof for Theorem 1(c)

In this subsection, we show $\text{var}(\chi^h(\mathcal{O}))$ is equal to the variance of unadjusted estimator τ_1^{UNADJ} . To see this point, note that if we let $h(X_i, X_j) = e(X_i)[1 - e(X_j)]$ and $\omega(X_i, X_j) = 1$, then $\hat{\tau}_1^h$ defined in (9) of main text is equivalent to τ_1^{UNADJ} defined in (3) of main text. From this viewpoint, the unadjusted estimator $\hat{\tau}_1^{\text{UNADJ}}$ also belongs to the same class of balancing weighting estimators as $\hat{\tau}_1^h$. Therefore, by using a similar argument as in Theorem 1(a), the influence function of $\hat{\tau}_1^{\text{UNADJ}}$ is

$$\phi^{\text{UNADJ}}(\mathcal{O}) = 2 \{q_1^{\text{UNADJ}}(\mathcal{O}) - \tau_1\},$$

where $q_1^{\text{UNADJ}}(\mathcal{O}) = \mathbb{E} q^{\text{UNADJ}}(\cdot, \mathcal{O}) = \frac{1}{2} \left[\frac{Z}{\pi} g_1(Y) + \frac{1-Z}{1-\pi} g_0(Y) \right]$ with $q^{\text{UNADJ}}(\mathcal{O}_i, \mathcal{O}_j) = \frac{1}{2} [Z_i(1 - Z_j) \mathbb{I}(Y_i > Y_j) + Z_j(1 - Z_i) \mathbb{I}(Y_i < Y_j)] / \left[\frac{1}{n(n-1)} \sum_{i=1}^n \sum_{j=1}^n Z_i(1 - Z_j) \right]$. Typically, for the unadjusted estimator $\hat{\tau}_1^{\text{UNADJ}}$, $\pi = e(X)$

and $\tau_1^h = \tau_1$, so $2\{q_1^{\text{UNADJ}}(\mathcal{O}) - \tau_1\} = \chi^h(\mathcal{O})$ and $\text{var}(\hat{\tau}_1^{\text{UNADJ}}) = \text{var}(\chi^h(\mathcal{O}))$. Therefore, according to the result in (A.5), we know both IPW and OW weighting estimators have no larger variance than the unadjusted estimator. By Theorem 1(c), the asymptotic variance of $n^{1/2}(\hat{\tau}_1^{\text{UNADJ}} - \tau_1)$ can be consistently estimated by

$$n^{-1} \sum_{i=1}^n [2\{\hat{q}_1^{\text{UNADJ}}(\mathcal{O}_i) - \hat{\tau}_1^{\text{UNADJ}}\}]^{\otimes 2}, \quad (\text{A.6})$$

where $\hat{q}_1^{\text{UNADJ}}(\mathcal{O}_i) = (n-1)^{-1} \sum_{j \neq i}^n q^{\text{UNADJ}}(\mathcal{O}_i, \mathcal{O}_j)$.

Web Appendix C: Closed-form variance estimation for $\widehat{\text{var}}(\hat{\tau}_1^{\text{IPW}})$ and $\widehat{\text{var}}(\hat{\tau}_1^{\text{OW}})$

According to the result in Theorem 1(a) of main text, $\text{var}(\hat{\tau}_1^h)$ can be consistently estimated as

$$n^{-2} \sum_{i=1}^n \left[2\{\hat{q}_1^h(\mathcal{O}_i; \hat{\beta}_n) - \hat{\tau}_1^h\} + \tilde{l}_\beta^\top(Z_i, X_i; \hat{\beta}) A_1^h \right]^{\otimes 2}, \quad (\text{A.7})$$

where $\hat{q}_1^h(\mathcal{O}_i; \hat{\beta}) = (n-1)^{-1} \sum_{j \neq i}^n q^h(\mathcal{O}_i, \mathcal{O}_j; \hat{\beta})$ and $A_1^h = \binom{n}{2}^{-1} \sum_{i=1}^n \sum_{j>i}^n \frac{\partial}{\partial \beta} q^h(\mathcal{O}_i, \mathcal{O}_j; \beta) |_{\beta=\hat{\beta}}$, $\tilde{l}_\beta(Z, X; \hat{\beta})$ is the influence function for $\hat{\beta}$ and $a^{\otimes 2} = aa^\top$ for any vector a . Therefore, for $\hat{\tau}_1^{\text{IPW}}$ (i.e., $\hat{\tau}_1^h$ with $\omega(X_i, X_j) = 1/\{\hat{e}(X_i)[1 - \hat{e}(X_j)]\}$) and $\hat{\tau}_1^{\text{OW}}$ (i.e., $\hat{\tau}_1^h$ with $\omega(X_i, X_j) = \hat{e}(X_j)[1 - \hat{e}(X_i)]$), their consistent variance estimators are given by

$$\widehat{\text{var}}(\hat{\tau}_1^{\text{IPW}}) = n^{-2} \sum_{i=1}^n \left[2\{\hat{q}_1^{\text{IPW}}(\mathcal{O}_i; \hat{\beta}) - \hat{\tau}_1^{\text{IPW}}\} + \tilde{l}_\beta^\top(Z_i, X_i; \hat{\beta}) A_1^{\text{IPW}} \right]^{\otimes 2} \quad (\text{A.8})$$

and

$$\widehat{\text{var}}(\hat{\tau}_1^{\text{OW}}) = n^{-2} \sum_{i=1}^n \left[2\{\hat{q}_1^{\text{OW}}(\mathcal{O}_i; \hat{\beta}_n) - \hat{\tau}_1^{\text{OW}}\} + \tilde{l}_\beta^\top(Z_i, X_i; \hat{\beta}) A_1^{\text{OW}} \right]^{\otimes 2} \quad (\text{A.9})$$

respectively, where

$$\begin{aligned} \hat{q}_1^{\text{IPW}}(\mathcal{O}_i; \hat{\beta}) &= \frac{1}{n-1} \sum_{j \neq i}^n q^{\text{IPW}}(\mathcal{O}_i, \mathcal{O}_j; \hat{\beta}), & \hat{q}_1^{\text{OW}}(\mathcal{O}_i; \hat{\beta}) &= \frac{1}{n-1} \sum_{j \neq i}^n q^{\text{OW}}(\mathcal{O}_i, \mathcal{O}_j; \hat{\beta}), \\ A_1^{\text{IPW}} &= \binom{n}{2}^{-1} \sum_{i=1}^n \sum_{j>i}^n \frac{\partial}{\partial \beta} q^{\text{IPW}}(\mathcal{O}_i, \mathcal{O}_j; \hat{\beta}), & A_1^{\text{OW}} &= \binom{n}{2}^{-1} \sum_{i=1}^n \sum_{j>i}^n \frac{\partial}{\partial \beta} q^{\text{OW}}(\mathcal{O}_i, \mathcal{O}_j; \hat{\beta}), \\ q^{\text{IPW}}(\mathcal{O}_i, \mathcal{O}_j; \hat{\beta}) &= \frac{1}{2} \left\{ \frac{Z_i(1-Z_j)\mathbb{I}(Y_i > Y_j)}{\hat{e}(X_i)[1 - \hat{e}(X_j)]} + \frac{Z_j(1-Z_i)\mathbb{I}(Y_i < Y_j)}{\hat{e}(X_j)[1 - \hat{e}(X_i)]} \right\} / \left\{ \frac{1}{n(n-1)} \sum_{i=1}^n \sum_{j=1}^n \frac{Z_i(1-Z_j)}{\hat{e}(X_i)[1 - \hat{e}(X_j)]} \right\}, \\ q^{\text{OW}}(\mathcal{O}_i, \mathcal{O}_j; \hat{\beta}) &= \frac{\hat{e}(X_j)[1 - \hat{e}(X_i)]Z_i(1-Z_j)\mathbb{I}(Y_i > Y_j) + \hat{e}(X_i)[1 - \hat{e}(X_j)]Z_j(1-Z_i)\mathbb{I}(Y_i < Y_j)}{2\{n^{-1}(n-1)^{-1} \sum_{i=1}^n \sum_{j=1}^n Z_i(1-Z_j)[1 - \hat{e}(X_i)]\hat{e}(X_j)\}}. \end{aligned}$$

Next, we provide explicit expressions for computing $\tilde{l}_\beta(Z, X; \hat{\beta})$, A_1^{IPW} and A_1^{OW} .

C1. Influence function for logistic regression

In this paper, we apply a logistic regression model to describe the propensity score, that is, $e(X_i) = e(X_i; \beta) = \text{Pr}(Z_i = 1|X_i) = \{1 + \exp(-X_i^\top \beta)\}^{-1}$ for $i = 1, \dots, n$. Then, the score function of $\hat{\beta}$ is

$$n^{-1/2} \sum_{i=1}^n X_i [Z_i - e(X_i; \hat{\beta})] = 0.$$

By using Taylor expansion, we expand the score equation around the true parameter β , i.e.,

$$\begin{aligned} 0 &= n^{-1/2} \sum_{i=1}^n X_i [Z_i - e(X_i; \hat{\beta})] \\ &= n^{-1/2} \sum_{i=1}^n X_i [Z_i - e(X_i; \beta)] + n^{-1/2} \frac{\partial \{\sum_{i=1}^n X_i [Z_i - e(X_i; \beta)]\}}{\partial \beta} (\hat{\beta} - \beta) + o_p(1) \\ &= n^{-1/2} \sum_{i=1}^n X_i [Z_i - e(X_i; \beta)] - \left[\frac{1}{n} \sum_{i=1}^n X_i X_i^\top e(X_i; \beta) \{1 - e(X_i; \beta)\} \right] \sqrt{n} (\hat{\beta} - \beta) + o_p(1), \end{aligned}$$

which leads to

$$\begin{aligned} \sqrt{n} (\hat{\beta} - \beta) &= \left[\frac{1}{n} \sum_{i=1}^n X_i X_i^\top e(X_i; \beta) \{1 - e(X_i; \beta)\} \right]^{-1} \frac{1}{\sqrt{n}} \sum_{i=1}^n X_i [Z_i - e(X_i; \beta)] + o_p(1) \\ &= E_{\beta\beta}^{-1} \frac{1}{\sqrt{n}} \sum_{i=1}^n X_i [Z_i - e(X_i; \beta)] + o_p(1), \end{aligned}$$

where $E_{\beta\beta} = \frac{1}{n} \sum_{i=1}^n X_i X_i^\top e(X_i; \beta) \{1 - e(X_i; \beta)\}$. Thus, according to Tsiatis (2006), the influence function $\tilde{l}_\beta(Z_i, X_i; \hat{\beta})$ is

$$\tilde{l}_\beta(Z_i, X_i; \hat{\beta}) = E_{\hat{\beta}\hat{\beta}}^{-1} X_i [Z_i - e(X_i; \hat{\beta})] = E_{\hat{\beta}\hat{\beta}}^{-1} X_i [Z_i - \hat{e}(X_i)]. \quad (\text{A.10})$$

C2. Expression for A_1^{IPW}

For $q^{\text{IPW}}(\mathcal{O}_i, \mathcal{O}_j; \hat{\beta})$ involving in A_1^{IPW} , conducting derivative of $q^{\text{IPW}}(\mathcal{O}_i, \mathcal{O}_j; \beta)$ with respect to β gives

$$\begin{aligned} &\frac{\partial}{\partial \beta} q^{\text{IPW}}(\mathcal{O}_i, \mathcal{O}_j; \beta) \\ &= \frac{1}{2} \left[\frac{0 - \{e(X_i)(1 - e(X_i))X_i(1 - e(X_j)) + e(X_i)[-e(X_j)(1 - e(X_j))X_j]\}Z_i(1 - Z_j)I(Y_i > Y_j)}{e^2(X_i)(1 - e(X_j))^2} \right. \\ &\quad \left. + \frac{0 - \{e(X_j)(1 - e(X_j))X_j(1 - e(X_i)) + e(X_j)[-e(X_i)(1 - e(X_i))X_i]\}Z_j(1 - Z_i)I(Y_i < Y_j)}{e^2(X_j)(1 - e(X_i))^2} \right] / \\ &\quad \left[\frac{1}{n(n-1)} \sum_{i=1}^n \sum_{j=1}^n \frac{Z_i(1 - Z_j)}{e(X_i)[1 - e(X_j)]} \right] - \frac{\frac{1}{2} \left[\frac{Z_i(1 - Z_j)I(Y_i > Y_j)}{e(X_i)(1 - e(X_j))} + \frac{Z_j(1 - Z_i)I(Y_i < Y_j)}{e(X_j)(1 - e(X_i))} \right]}{\left[\frac{1}{n(n-1)} \sum_{i=1}^n \sum_{j=1}^n \frac{Z_i(1 - Z_j)}{e(X_i)[1 - e(X_j)]} \right]^2} \times \\ &\quad \left[\frac{1}{n(n-1)} \sum_{i=1}^n \sum_{j=1}^n \frac{Z_i(1 - Z_j)[e(X_j)X_j - (1 - e(X_i))X_i]}{e(X_i)[1 - e(X_j)]} \right] \\ &= A^{\text{IPW}}(\mathcal{O}_i, \mathcal{O}_j; \beta) / C^{\text{IPW}} - q^{\text{IPW}}(\mathcal{O}_i, \mathcal{O}_j; \beta) \times B^{\text{IPW}} / C^{\text{IPW}}, \quad (\text{A.11}) \end{aligned}$$

where

$$\begin{aligned}
A^{\text{IPW}}(\mathcal{O}_i, \mathcal{O}_j; \beta) &= \frac{1}{2} \left[\frac{[e(X_j)X_j - (1 - e(X_i))X_i]Z_i(1 - Z_j)I(Y_i > Y_j)}{e(X_i)(1 - e(X_j))} \right. \\
&\quad \left. + \frac{[e(X_i)X_i - (1 - e(X_j))X_j]Z_j(1 - Z_i)I(Y_i < Y_j)}{e(X_j)(1 - e(X_i))} \right], \\
B^{\text{IPW}} &= \frac{1}{n(n-1)} \sum_{i=1}^n \sum_{j=1}^n \frac{Z_i(1 - Z_j)[e(X_j)X_j - (1 - e(X_i))X_i]}{e(X_i)[1 - e(X_j)]}, \\
C^{\text{IPW}} &= \frac{1}{n(n-1)} \sum_{i=1}^n \sum_{j=1}^n \frac{Z_i(1 - Z_j)}{e(X_i)[1 - e(X_j)]}.
\end{aligned}$$

Therefore, A_1^{IPW} can be obtained by putting (A.11) into it.

Remark 1

1. C^{IPW} is a numerical value and B^{IPW} is a vector once we have observed data and obtain $\hat{\beta}$.
2. A_1^{IPW} can be further simplified as followings:

$$\begin{aligned}
A_1^{\text{IPW}} &= \frac{1}{\hat{C}^{\text{IPW}}} \left[\frac{1}{n(n-1)} \sum_{i=1}^n \sum_{j=1}^n \frac{[\hat{e}(X_j)X_j - (1 - \hat{e}(X_i))X_i]Z_i(1 - Z_j)\mathbb{I}(Y_i > Y_j)}{\hat{e}(X_i)(1 - \hat{e}(X_j))} \right] - \frac{\hat{B}^{\text{IPW}}}{\hat{C}^{\text{IPW}}} \hat{\tau}_1^{\text{IPW}} \\
&= \frac{\sum_{i=1}^n \sum_{j=1}^n [\hat{e}(X_j)X_j - (1 - \hat{e}(X_i))X_i] \omega(X_i, X_j) Z_i(1 - Z_j) \mathbb{I}(Y_i > Y_j)}{\sum_{i=1}^n \sum_{j=1}^n Z_i(1 - Z_j) / [\hat{e}(X_i)(1 - \hat{e}(X_j))]} - \frac{\hat{B}^{\text{IPW}}}{\hat{C}^{\text{IPW}}} \hat{\tau}_1^{\text{IPW}},
\end{aligned}$$

where $\omega(X_i, X_j) = \frac{1}{\hat{e}(X_i)(1 - \hat{e}(X_j))}$.

C3. Expression for A_1^{OW}

Next, we show how to compute $\frac{\partial}{\partial \beta} q^{\text{OW}}(\mathcal{O}_i, \mathcal{O}_j; \beta)$ involving in A_1^{OW} . Conducting derivative of $q^{\text{OW}}(\mathcal{O}_i, \mathcal{O}_j; \hat{\beta})$ with respect to β , we have

$$\begin{aligned}
\frac{\partial}{\partial \beta} q^{\text{OW}}(\mathcal{O}_i, \mathcal{O}_j; \beta) &= \frac{\frac{1}{2}[(1 - e(X_j))X_j - e(X_i)X_i]\omega(X_i, X_j)Z_i(1 - Z_j)I(Y_i > Y_j)}{\frac{1}{n(n-1)} \sum_{i=1}^n \sum_{j=1}^n Z_i(1 - Z_j)(1 - e(X_i))e(X_j)} \\
&\quad + \frac{\frac{1}{2}[(1 - e(X_i))X_i - e(X_j)X_j]\omega(X_j, X_i)Z_j(1 - Z_i)I(Y_i < Y_j)}{\frac{1}{n(n-1)} \sum_{i=1}^n \sum_{j=1}^n Z_i(1 - Z_j)(1 - e(X_i))e(X_j)} \\
&\quad - \frac{\frac{1}{2}[\omega(X_i, X_j)Z_i(1 - Z_j)I(Y_i > Y_j) + \omega(X_j, X_i)Z_j(1 - Z_i)I(Y_i < Y_j)]}{\left[\frac{1}{n(n-1)} \sum_{i=1}^n \sum_{j=1}^n Z_i(1 - Z_j)(1 - e(X_i))e(X_j) \right]^2} \times \\
&\quad \left[\frac{1}{n(n-1)} \sum_{i=1}^n \sum_{j=1}^n Z_i(1 - Z_j)(1 - e(X_i))e(X_j)[(1 - e(X_j))X_j - e(X_i)X_i] \right] \\
&= A^{\text{OW}}(\mathcal{O}_i, \mathcal{O}_j; \beta) / C^{\text{OW}} - q^{\text{OW}}(\mathcal{O}_i, \mathcal{O}_j; \beta) \times B^{\text{OW}} / C^{\text{OW}}, \tag{A.12}
\end{aligned}$$

where

$$\begin{aligned}
A^{\text{OW}}(\mathcal{O}_i, \mathcal{O}_j; \beta) &= \frac{1}{2} \{ [(1 - e(X_j))X_j - e(X_i)X_i] \omega(X_i, X_j) Z_i(1 - Z_j) I(Y_i > Y_j) \\
&\quad + [(1 - e(X_i))X_i - e(X_j)X_j] \omega(X_j, X_i) Z_j(1 - Z_i) I(Y_i < Y_j) \}, \\
B^{\text{OW}} &= \frac{1}{n(n-1)} \sum_{i=1}^n \sum_{j=1}^n Z_i(1 - Z_j)(1 - e(X_i))e(X_j)[(1 - e(X_j))X_j - e(X_i)X_i], \\
C^{\text{OW}} &= \frac{1}{n(n-1)} \sum_{i=1}^n \sum_{j=1}^n Z_i(1 - Z_j)(1 - e(X_i))e(X_j).
\end{aligned}$$

Therefore, A_1^{OW} can be obtained by putting (A.12) into it.

Remark 2

1. C^{OW} is a numerical value and B^{OW} is a vector once we have observed data and obtain $\hat{\beta}$.
2. A_1^{OW} can be further simplified as followings:

$$\begin{aligned}
A_1^{\text{OW}} &= \frac{1}{\hat{C}^{\text{OW}}} \left[\frac{1}{n(n-1)} \sum_{i=1}^n \sum_{j=1}^n [(1 - \hat{e}(X_j))X_j - \hat{e}(X_i)X_i] \omega(X_i, X_j) Z_i(1 - Z_j) \mathbb{I}(Y_i > Y_j) \right] - \frac{\hat{B}^{\text{OW}}}{\hat{C}^{\text{OW}}} \hat{\tau}_1^{\text{OW}} \\
&= \frac{\sum_{i=1}^n \sum_{j=1}^n [(1 - \hat{e}(X_j))X_j - \hat{e}(X_i)X_i] \omega(X_i, X_j) Z_i(1 - Z_j) \mathbb{I}(Y_i > Y_j)}{\sum_{i=1}^n \sum_{j=1}^n Z_i(1 - Z_j)(1 - \hat{e}(X_i))\hat{e}(X_j)} - \frac{\hat{B}^{\text{OW}}}{\hat{C}^{\text{OW}}} \hat{\tau}_1^{\text{OW}},
\end{aligned}$$

where $\omega(X_i, X_j) = (1 - \hat{e}(X_i))\hat{e}(X_j)$.

Web Appendix D: Closed-form variance estimation for $\widehat{\text{var}}(\hat{\tau}_1^{\text{AIPW}})$ and $\widehat{\text{var}}(\hat{\tau}_1^{\text{AOW}})$

Note that based on Theorem 2 of main text, the closed-form variance estimators of $\hat{\tau}_1^{\text{AIPW}}$ and $\hat{\tau}_1^{\text{AOW}}$ are

$$\widehat{\text{var}}(\hat{\tau}_1^{\text{AIPW}}) = n^{-2} \sum_{i=1}^n \left[2\{\hat{r}_1^{\text{AIPW}}(\mathcal{O}_i; \hat{\beta}, \hat{\alpha}) - \hat{\tau}_1^{\text{AIPW}}\} + \tilde{l}_\beta^\top(Z_i, X_i; \hat{\beta}) B_1^{\text{AIPW}} + \tilde{l}_\alpha^\top(Y_i, X_i; \hat{\alpha}) C_1^{\text{AIPW}} \right]^{\otimes 2}, \quad (\text{A.13})$$

and

$$\widehat{\text{var}}(\hat{\tau}_1^{\text{AOW}}) = n^{-2} \sum_{i=1}^n \left[2\{\hat{r}_1^{\text{AOW}}(\mathcal{O}_i; \hat{\beta}, \hat{\alpha}) - \hat{\tau}_1^{\text{AOW}}\} + \tilde{l}_\beta^\top(Z_i, X_i; \hat{\beta}) B_1^{\text{AOW}} + \tilde{l}_\alpha^\top(Y_i, X_i; \hat{\alpha}) C_1^{\text{AOW}} \right]^{\otimes 2}, \quad (\text{A.14})$$

respectively, where

$$\begin{aligned}
\hat{r}_1^{\text{AIPW}}(\mathcal{O}_i; \hat{\beta}, \hat{\alpha}) &= (n-1)^{-1} \sum_{j \neq i}^n r^{\text{AIPW}}(\mathcal{O}_i, \mathcal{O}_j; \hat{\beta}, \hat{\alpha}), & \hat{r}_1^{\text{AOW}}(\mathcal{O}_i; \hat{\beta}, \hat{\alpha}) &= (n-1)^{-1} \sum_{j \neq i}^n r^{\text{AOW}}(\mathcal{O}_i, \mathcal{O}_j; \hat{\beta}, \hat{\alpha}), \\
B_1^{\text{AIPW}} &= \binom{n}{2}^{-1} \sum_{i=1}^n \sum_{j>i}^n \frac{\partial}{\partial \beta} r^{\text{AIPW}}(\mathcal{O}_i, \mathcal{O}_j; \beta, \hat{\alpha})|_{\beta=\hat{\beta}}, & B_1^{\text{AOW}} &= \binom{n}{2}^{-1} \sum_{i=1}^n \sum_{j>i}^n \frac{\partial}{\partial \beta} r^{\text{AOW}}(\mathcal{O}_i, \mathcal{O}_j; \beta, \hat{\alpha})|_{\beta=\hat{\beta}}, \\
C_1^{\text{AIPW}} &= \binom{n}{2}^{-1} \sum_{i=1}^n \sum_{j>i}^n \frac{\partial}{\partial \alpha} r^{\text{AIPW}}(\mathcal{O}_i, \mathcal{O}_j; \hat{\beta}, \alpha)|_{\alpha=\hat{\alpha}}, & C_1^{\text{AOW}} &= \binom{n}{2}^{-1} \sum_{i=1}^n \sum_{j>i}^n \frac{\partial}{\partial \alpha} r^{\text{AOW}}(\mathcal{O}_i, \mathcal{O}_j; \hat{\beta}, \alpha)|_{\alpha=\hat{\alpha}},
\end{aligned}$$

$$r^{\text{AIPW}}(\cdot, \cdot; \hat{\beta}, \hat{\alpha}) = \frac{\frac{1}{2} \left\{ \frac{Z_i(1-Z_j)[\mathbb{I}(Y_i > Y_j) - \hat{\mu}(X_i, X_j)]}{\hat{e}(X_i)(1-\hat{e}(X_j))} + \frac{Z_j(1-Z_i)[\mathbb{I}(Y_j > Y_i) - \hat{\mu}(X_j, X_i)]}{\hat{e}(X_j)(1-\hat{e}(X_i))} \right\}}{\frac{1}{n(n-1)} \sum_{i=1}^n \sum_{j \neq i}^n Z_i(1-Z_j)/[\hat{e}(X_i)(1-\hat{e}(X_j))]} + \frac{1}{2} [\hat{\mu}(X_i, X_j) + \hat{\mu}(X_j, X_i)],$$

$$\begin{aligned} r^{\text{AOW}}(\cdot, \cdot; \hat{\beta}, \hat{\alpha}) &= \frac{\frac{1}{2} Z_i(1-Z_j)(1-\hat{e}(X_i))\hat{e}(X_j)[\mathbb{I}(Y_i > Y_j) - \hat{\mu}(X_i, X_j)]}{\frac{1}{n(n-1)} \sum_{i=1}^n \sum_{j \neq i}^n Z_i(1-Z_j)(1-\hat{e}(X_i))\hat{e}(X_j)} \\ &+ \frac{\frac{1}{2} Z_j(1-Z_i)(1-\hat{e}(X_j))\hat{e}(X_i)[\mathbb{I}(Y_j > Y_i) - \hat{\mu}(X_j, X_i)]}{\frac{1}{n(n-1)} \sum_{i=1}^n \sum_{j \neq i}^n Z_i(1-Z_j)(1-\hat{e}(X_i))\hat{e}(X_j)} \\ &+ \frac{\frac{1}{2} \hat{e}(X_i)(1-\hat{e}(X_i))\hat{e}(X_j)(1-\hat{e}(X_j))[\hat{\mu}(X_i, X_j) + \hat{\mu}(X_j, X_i)]}{\frac{1}{n(n-1)} \sum_{i=1}^n \sum_{j \neq i}^n \hat{e}(X_i)\hat{e}(X_j)(1-\hat{e}(X_i))(1-\hat{e}(X_j))}. \end{aligned}$$

Next, we provide explicit expressions for computing $\tilde{l}_\alpha(Y, X; \hat{\alpha})$, B_1^{AIPW} , C_1^{AIPW} , B_1^{AOW} and C_1^{AOW} .

D1. Inference for ordinal logistic regression

According to (A.13) and (A.14), to obtain $\hat{\tau}_1^{\text{AIPW}}$ (or $\hat{\tau}_1^{\text{AOW}}$) and its variance, we also need to compute $\hat{\alpha}$ and $\hat{\mu}(\cdot, \cdot)$ in addition to $\hat{\beta}$ and $\hat{e}(\cdot)$. In this paper, we apply ordinal logistic regression to model $\mu(\cdot, \cdot; \alpha)$ and use polr program in R package MASS to fit that. Note that in polr program, the ordinal logistic regression model for ordinal outcome with L categories is parameterized as

$$\text{logit}(P(Y \leq l)) = \alpha_{l0} - \xi^\top X, \quad (\text{A.15})$$

where $l = 1, 2, \dots, L-1$ and $\xi = (\xi_1, \dots, \xi_p)^\top$. Then based on (A.15), we have

$$\log \frac{P(Y \leq l)}{1 - P(Y \leq l)} = \alpha_{l0} - \xi^\top X \Leftrightarrow P(Y \leq l) = \frac{\exp(\alpha_{l0} - \xi^\top X)}{1 + \exp(\alpha_{l0} - \xi^\top X)}. \quad (\text{A.16})$$

Denote $\alpha = (\alpha_{10}, \dots, \alpha_{L-1,0}, \xi^\top)^\top$, $c_{il} = c_l(X_i; \alpha_{l0}, \xi) = P(Y_i \leq l | X_i) = \frac{\exp(\alpha_{l0} - \xi^\top X_i)}{1 + \exp(\alpha_{l0} - \xi^\top X_i)}$, $p_{il} = p_l(X_i; \alpha_{l0}, \xi) = P(Y_i = l | X_i)$, $\delta_{il} = 1$ if $Y_i = l$ and $\delta_{il} = 0$ if $Y_i \neq l$, for $i = 1, 2, \dots, n$ and $l = 1, 2, \dots, L$. Then the likelihood and log likelihood function for model (A.16) are

$$L_n(\alpha) = \prod_{i=1}^n \prod_{l=1}^L p_{il}^{\delta_{il}}, \quad \text{and} \quad l_n(\alpha) = \sum_{i=1}^n \sum_{l=1}^L \delta_{il} \log(p_{il}), \quad (\text{A.17})$$

respectively, where (1) $p_{i1} = c_{i1} = P(Y_i \leq 1 | X_i) = P(Y_i = 1 | X_i)$ if $Y_i = 1$; (2) $p_{il} = c_{il} - c_{i,l-1} = P(Y_i \leq l | X_i) - P(Y_i \leq l-1 | X_i)$ if $1 < Y_i = l < L$; (3) $p_{iL} = 1 - c_{i,L-1} = 1 - P(Y_i \leq L-1 | X_i)$ if $Y_i = L$.

Note that we consider $L = 3$ in our simulation, so we use $L = 3$ as an example to show how to obtain influence function of α . Influence function of α for other values of L can be obtained similarly. When $L = 3$, then the log-likelihood function (A.17) becomes

$$\begin{aligned} l_n(\alpha) &= \sum_{i=1}^n \sum_{l=1}^3 \delta_{il} \log(p_{il}) = \sum_{i=1}^n [\delta_{i1} \log p_{i1} + \delta_{i2} \log p_{i2} + \delta_{i3} \log p_{i3}] \\ &= \sum_{i=1}^n [\delta_{i1} \log c_{i1} + \delta_{i2} \log(c_{i2} - c_{i1}) + \delta_{i3} \log(1 - c_{i2})], \end{aligned}$$

where $\alpha = (\alpha_{10}, \alpha_{20}, \xi^\top)^\top$. It is noticed that $c_{i1} = \frac{\exp(\alpha_{10} - \xi^\top X_i)}{1 + \exp(\alpha_{10} - \xi^\top X_i)}$ and $c_{i2} = \frac{\exp(\alpha_{20} - \xi^\top X_i)}{1 + \exp(\alpha_{20} - \xi^\top X_i)}$, then $c_{i2} - c_{i1} = \frac{\exp(\alpha_{20} - \xi^\top X_i) - \exp(\alpha_{10} - \xi^\top X_i)}{(1 + \exp(\alpha_{20} - \xi^\top X_i))(1 + \exp(\alpha_{10} - \xi^\top X_i))}$. Therefore, the log-likelihood function $l_n(\alpha)$ can be rewritten as

$$\begin{aligned} l_n(\alpha) &= \sum_{i=1}^n \{ \delta_{i1}(\alpha_{10} - \xi^\top X_i) - \delta_{i1} \log[1 + \exp(\alpha_{10} - \xi^\top X_i)] + \delta_{i2} \log[\exp(\alpha_{20} - \xi^\top X_i) - \exp(\alpha_{10} - \xi^\top X_i)] \\ &\quad - \delta_{i2} \log[1 + \exp(\alpha_{10} - \xi^\top X_i)] - \delta_{i2} \log[1 + \exp(\alpha_{20} - \xi^\top X_i)] - \delta_{i3} \log[1 + \exp(\alpha_{20} - \xi^\top X_i)] \} \\ &= \sum_{i=1}^n \{ \delta_{i1}(\alpha_{10} - \xi^\top X_i) - (\delta_{i1} + \delta_{i2}) \log[1 + \exp(\alpha_{10} - \xi^\top X_i)] - (\delta_{i2} + \delta_{i3}) \log[1 + \exp(\alpha_{20} - \xi^\top X_i)] \\ &\quad + \delta_{i2} \log[\exp(\alpha_{20} - \xi^\top X_i) - \exp(\alpha_{10} - \xi^\top X_i)] \}. \end{aligned}$$

Thus, conducting derivative of $l_n(\alpha)$ with respect to α , we have

$$\begin{aligned} \frac{\partial l_n(\alpha)}{\partial \alpha} &= \sum_{i=1}^n \left[\begin{array}{c} \delta_{i1} - (\delta_{i1} + \delta_{i2}) \frac{\exp(\alpha_{10} - \xi^\top X_i)}{1 + \exp(\alpha_{10} - \xi^\top X_i)} - \delta_{i2} \frac{\exp(\alpha_{10} - \xi^\top X_i)}{\exp(\alpha_{20} - \xi^\top X_i) - \exp(\alpha_{10} - \xi^\top X_i)} \\ - (\delta_{i2} + \delta_{i3}) \frac{\exp(\alpha_{20} - \xi^\top X_i)}{1 + \exp(\alpha_{20} - \xi^\top X_i)} + \delta_{i2} \frac{\exp(\alpha_{20} - \xi^\top X_i)}{\exp(\alpha_{20} - \xi^\top X_i) - \exp(\alpha_{10} - \xi^\top X_i)} \\ - \delta_{i1} X_i + (\delta_{i1} + \delta_{i2}) \frac{\exp(\alpha_{10} - \xi^\top X_i)}{1 + \exp(\alpha_{10} - \xi^\top X_i)} X_i + (\delta_{i2} + \delta_{i3}) \frac{\exp(\alpha_{20} - \xi^\top X_i)}{1 + \exp(\alpha_{20} - \xi^\top X_i)} X_i - \delta_{i2} \frac{\exp(\alpha_{20} - \xi^\top X_i) - \exp(\alpha_{10} - \xi^\top X_i)}{\exp(\alpha_{20} - \xi^\top X_i) - \exp(\alpha_{10} - \xi^\top X_i)} X_i \end{array} \right] \\ &= \sum_{i=1}^n \left[\begin{array}{c} \delta_{i1} - (\delta_{i1} + \delta_{i2}) c_{i1} - \delta_{i2} \frac{1}{\exp(\alpha_{20} - \alpha_{10}) - 1} \\ - (\delta_{i2} + \delta_{i3}) c_{i2} + \delta_{i2} \frac{1}{1 - \exp(\alpha_{10} - \alpha_{20})} \\ \{ -(\delta_{i1} + \delta_{i2}) + (\delta_{i1} + \delta_{i2}) c_{i1} + (\delta_{i2} + \delta_{i3}) c_{i2} \} X_i \end{array} \right]. \end{aligned}$$

Then, the score function of α is

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n \left[\begin{array}{c} \delta_{i1} - (\delta_{i1} + \delta_{i2}) \hat{c}_{i1} - \delta_{i2} \frac{1}{\exp(\hat{\alpha}_{20} - \hat{\alpha}_{10}) - 1} \\ - (\delta_{i2} + \delta_{i3}) \hat{c}_{i2} + \delta_{i2} \frac{1}{1 - \exp(\hat{\alpha}_{10} - \hat{\alpha}_{20})} \\ \{ -(\delta_{i1} + \delta_{i2}) + (\delta_{i1} + \delta_{i2}) \hat{c}_{i1} + (\delta_{i2} + \delta_{i3}) \hat{c}_{i2} \} X_i \end{array} \right] = 0.$$

Next, we expand the above score equation around the true parameter α , which leads to

$$0 = \frac{1}{\sqrt{n}} \sum_{i=1}^n \left[\begin{array}{c} \delta_{i1} - (\delta_{i1} + \delta_{i2}) c_{i1} - \delta_{i2} \frac{1}{\exp(\alpha_{20} - \alpha_{10}) - 1} \\ - (\delta_{i2} + \delta_{i3}) c_{i2} + \delta_{i2} \frac{1}{1 - \exp(\alpha_{10} - \alpha_{20})} \\ \{ -(\delta_{i1} + \delta_{i2}) + (\delta_{i1} + \delta_{i2}) c_{i1} + (\delta_{i2} + \delta_{i3}) c_{i2} \} X_i \end{array} \right] + \frac{1}{\sqrt{n}} \sum_{i=1}^n \left[\begin{array}{ccc} F_{i11} & F_{i12} & F_{i13} \\ F_{i21} & F_{i22} & F_{i23} \\ F_{i31} & F_{i32} & F_{i33} \end{array} \right] (\hat{\alpha} - \alpha) + o_p(1),$$

where

$$\begin{aligned} F_{i11} &= -(\delta_{i1} + \delta_{i2})(c_{i1} - c_{i1}^2) - \delta_{i2} \frac{e^{\alpha_{20} - \alpha_{10}}}{(e^{\alpha_{20} - \alpha_{10}} - 1)^2}, \\ F_{i12} &= F_{i21} = \delta_{i2} \frac{e^{\alpha_{20} - \alpha_{10}}}{(e^{\alpha_{20} - \alpha_{10}} - 1)^2}, \\ F_{i13} &= F_{i31}^\top = (\delta_{i1} + \delta_{i2})(c_{i1} - c_{i1}^2) X_i^\top, \\ F_{i22} &= -(\delta_{i2} + \delta_{i3})(c_{i2} - c_{i2}^2) - \delta_{i2} \frac{e^{\alpha_{10} - \alpha_{20}}}{(1 - e^{\alpha_{10} - \alpha_{20}})^2}, \\ F_{i23} &= F_{i32}^\top = (\delta_{i2} + \delta_{i3})(c_{i2} - c_{i2}^2) X_i^\top, \\ F_{i33} &= -\{(\delta_{i1} + \delta_{i2})(c_{i1} - c_{i1}^2) + (\delta_{i2} + \delta_{i3})(c_{i2} - c_{i2}^2)\} X_i X_i^\top. \end{aligned}$$

Therefore,

$$\sqrt{n}(\hat{\alpha} - \alpha) = E_{\alpha\alpha}^{-1} \frac{1}{\sqrt{n}} \sum_{i=1}^n \left[\begin{array}{c} \delta_{i1} - (\delta_{i1} + \delta_{i2}) c_{i1} - \delta_{i2} \frac{1}{\exp(\alpha_{20} - \alpha_{10}) - 1} \\ - (\delta_{i2} + \delta_{i3}) c_{i2} + \delta_{i2} \frac{1}{1 - \exp(\alpha_{10} - \alpha_{20})} \\ \{ -(\delta_{i1} + \delta_{i2}) + (\delta_{i1} + \delta_{i2}) c_{i1} + (\delta_{i2} + \delta_{i3}) c_{i2} \} X_i \end{array} \right] + o_p(1),$$

where $E_{\alpha\alpha} = -\frac{1}{n} \sum_{i=1}^n \begin{bmatrix} F_{i11} & F_{i12} & F_{i13} \\ F_{i21} & F_{i22} & F_{i23} \\ F_{i31} & F_{i32} & F_{i33} \end{bmatrix}$. According to Tsiatis (2006), the influence function $\tilde{l}_\alpha(Y_i, X_i; \hat{\alpha})$ is

$$\begin{aligned} \tilde{l}_\alpha(Y_i, X_i; \hat{\alpha}) &= E_{\hat{\alpha}\hat{\alpha}}^{-1} \begin{bmatrix} \delta_{i1} - (\delta_{i1} + \delta_{i2})\hat{c}_{i1} - \delta_{i2} \frac{1}{\exp(\hat{\alpha}_{20} - \hat{\alpha}_{10}) - 1} \\ -(\delta_{i2} + \delta_{i3})\hat{c}_{i2} + \delta_{i2} \frac{1}{1 - \exp(\hat{\alpha}_{10} - \hat{\alpha}_{20})} \\ \{-(\delta_{i1} + \delta_{i2}) + (\delta_{i1} + \delta_{i2})\hat{c}_{i1} + (\delta_{i2} + \delta_{i3})\hat{c}_{i2}\}X_i \end{bmatrix} \\ &= E_{\hat{\alpha}\hat{\alpha}}^{-1} \begin{bmatrix} \delta_{i1} - (\delta_{i1} + \delta_{i2})c_1(X_i; \hat{\alpha}_{10}, \hat{\xi}) - \delta_{i2} \frac{1}{\exp(\hat{\alpha}_{20} - \hat{\alpha}_{10}) - 1} \\ -(\delta_{i2} + \delta_{i3})c_2(X_i; \hat{\alpha}_{20}, \hat{\xi}) + \delta_{i2} \frac{1}{1 - \exp(\hat{\alpha}_{10} - \hat{\alpha}_{20})} \\ \{-(\delta_{i1} + \delta_{i2}) + (\delta_{i1} + \delta_{i2})c_1(X_i; \hat{\alpha}_{10}, \hat{\xi}) + (\delta_{i2} + \delta_{i3})c_2(X_i; \hat{\alpha}_{20}, \hat{\xi})\}X_i \end{bmatrix} \end{aligned} \quad (\text{A.18})$$

Specifically, for win statistics of ordinal outcomes, $p_l^z(X; \alpha)$ is denoted as the model based conditional probability of $Y(z) = l$ given X , where $z \in \{0, 1\}$. So, $p_l^z(X_i; \alpha) = p_{il}$ for z group. Assuming i is from $z = 1$ group and j is from $z = 0$ group, then,

$$\begin{aligned} \mu(X_i, X_j; \alpha) &= E[\mathbb{I}(Y_i(1) > Y_j(0))] = \sum_{l=1}^3 \sum_{l'=1}^{l-1} p_l^1(X_i; \alpha) p_{l'}^0(X_j; \alpha) \\ &= p_2^1(X_i; \alpha) p_1^0(X_j; \alpha) + p_3^1(X_i; \alpha) p_1^0(X_j; \alpha) + p_3^1(X_i; \alpha) p_2^0(X_j; \alpha) \\ &= (c_{i2} - c_{i1})c_{j1} + (1 - c_{i2})c_{j1} + (1 - c_{i2})(c_{j2} - c_{j1}) \\ &= (1 - c_{i1})c_{j1} + (1 - c_{i2})(c_{j2} - c_{j1}), \\ \mu(X_j, X_i; \alpha) &= E[\mathbb{I}(Y_i(1) < Y_j(0))] = \sum_{l=1}^3 \sum_{l'=l+1}^3 p_l^1(X_i; \alpha) p_{l'}^0(X_j; \alpha) \\ &= p_1^1(X_i; \alpha) p_2^0(X_j; \alpha) + p_1^1(X_i; \alpha) p_3^0(X_j; \alpha) + p_2^1(X_i; \alpha) p_3^0(X_j; \alpha) \\ &= c_{i1}(c_{j2} - c_{j1}) + c_{i1}(1 - c_{j2}) + (c_{i2} - c_{i1})(1 - c_{j2}) \\ &= (1 - c_{j1})c_{i1} + (1 - c_{j2})(c_{i2} - c_{i1}). \end{aligned}$$

Thus, the derivative of $\mu(X_i, X_j; \alpha)$ and $\mu(X_i, X_j; \alpha)$ with respect to α is

$$\begin{aligned} \tilde{\mu}(X_i, X_j; \alpha) &= \frac{\partial \mu(X_i, X_j; \alpha)}{\partial \alpha} \\ &= \begin{bmatrix} -c_{i1}(1 - c_{i1})c_{j1} + c_{j1}(1 - c_{j1})(c_{i2} - c_{i1}) \\ -c_{i2}(1 - c_{i2})(c_{j2} - c_{j1}) + (1 - c_{i2})c_{j2}(1 - c_{j2}) \\ \{c_{j1}(c_{i1} - c_{i1}^2) + (c_{i2} - c_{i2}^2)(c_{j2} - c_{j1})\}X_i - \{(c_{i2} - c_{i1})(c_{j1} - c_{j1}^2) + (1 - c_{i2})(c_{j2} - c_{j2}^2)\}X_j \end{bmatrix}, \\ \tilde{\mu}(X_j, X_i; \alpha) &= \frac{\partial \mu(X_j, X_i; \alpha)}{\partial \alpha} \\ &= \begin{bmatrix} -c_{j1}(1 - c_{j1})c_{i1} + c_{i1}(1 - c_{i1})(c_{j2} - c_{j1}) \\ -c_{j2}(1 - c_{j2})(c_{i2} - c_{i1}) + (1 - c_{j2})c_{i2}(1 - c_{i2}) \\ \{c_{i1}(c_{j1} - c_{j1}^2) + (c_{j2} - c_{j2}^2)(c_{i2} - c_{i1})\}X_j - \{(c_{j2} - c_{j1})(c_{i1} - c_{i1}^2) + (1 - c_{j2})(c_{i2} - c_{i2}^2)\}X_i \end{bmatrix}, \end{aligned} \quad (\text{A.19})$$

where c_{i1} , c_{i2} , c_{j1} and c_{j2} are defined in (A.17).

D2. Expressions for B_1^{AIPW} and C_1^{AIPW}

In main text, we have claimed that asymptotic variance of $\hat{\tau}_1^{\text{AIPW}}$ can be obtained by Theorem 2. Specifically, when $\omega(X_i, X_j) = \frac{1}{\hat{e}(X_i)(1-\hat{e}(X_j))}$, $\hat{\tau}_1^{\text{AW}} = \hat{\tau}_1^{\text{AIPW}}$, and relative detailed expressions are as follows:

$$\begin{aligned}
\hat{\tau}_1^{\text{AIPW}} &= \binom{n}{2}^{-1} \sum_{i=1}^n \sum_{j>i}^n r^{\text{AIPW}}(\cdot, \cdot; \hat{\beta}, \hat{\alpha}), \\
r^{\text{AIPW}}(\cdot, \cdot; \hat{\beta}, \hat{\alpha}) &= \frac{\frac{1}{2} \left\{ \frac{Z_i(1-Z_j)[\mathbb{I}(Y_i > Y_j) - \hat{\mu}(X_i, X_j)]}{\hat{e}(X_i)(1-\hat{e}(X_j))} + \frac{Z_j(1-Z_i)[\mathbb{I}(Y_j > Y_i) - \hat{\mu}(X_j, X_i)]}{\hat{e}(X_j)(1-\hat{e}(X_i))} \right\}}{\frac{1}{n(n-1)} \sum_{i=1}^n \sum_{j \neq i}^n Z_i(1-Z_j)/[\hat{e}(X_i)(1-\hat{e}(X_j))]} \\
&\quad + \frac{1}{2} [\hat{\mu}(X_i, X_j) + \hat{\mu}(X_j, X_i)], \\
\hat{\mu}(X_i, X_j) &= \mu(X_i, X_j; \hat{\alpha}) = (1 - \hat{c}_{i1})\hat{c}_{j1} + (1 - \hat{c}_{i2})(\hat{c}_{j2} - \hat{c}_{j1}), \\
\hat{\mu}(X_j, X_i) &= \mu(X_j, X_i; \hat{\alpha}) = (1 - \hat{c}_{j1})\hat{c}_{i1} + (1 - \hat{c}_{j2})(\hat{c}_{i2} - \hat{c}_{i1}), \\
\hat{\tau}_1^{\text{AIPW}}(\mathcal{O}_i; \hat{\beta}, \hat{\alpha}) &= (n-1)^{-1} \sum_{j \neq i}^n r^{\text{AIPW}}(\cdot, \cdot; \hat{\beta}, \hat{\alpha}), \\
B_1^{\text{AIPW}} &= \tilde{B}_1^{\text{AIPW}} - \tilde{B}_3^{\text{AIPW}} * \frac{\tilde{B}_2^{\text{AIPW}}}{\tilde{B}_4^{\text{AIPW}}}, \\
\tilde{B}_1^{\text{AIPW}} &= \frac{\sum_{i=1}^n \sum_{j \neq i}^n \frac{[\hat{e}(X_j)X_j - (1-\hat{e}(X_i))X_i]Z_i(1-Z_j)[\mathbb{I}(Y_i > Y_j) - \hat{\mu}(X_i, X_j)]}{[\hat{e}(X_i)(1-\hat{e}(X_j))]}}{\sum_{i=1}^n \sum_{j \neq i}^n Z_i(1-Z_j)/[\hat{e}(X_i)(1-\hat{e}(X_j))]}, \\
\tilde{B}_2^{\text{AIPW}} &= \frac{\sum_{i=1}^n \sum_{j \neq i}^n Z_i(1-Z_j)[\mathbb{I}(Y_i > Y_j) - \hat{\mu}(X_i, X_j)]/[\hat{e}(X_i)(1-\hat{e}(X_j))]}{\sum_{i=1}^n \sum_{j \neq i}^n Z_i(1-Z_j)/[\hat{e}(X_i)(1-\hat{e}(X_j))]}, \\
\tilde{B}_3^{\text{AIPW}} &= \sum_{i=1}^n \sum_{j \neq i}^n \frac{Z_i(1-Z_j)[\hat{e}(X_j)X_j - (1-\hat{e}(X_i))X_i]}{\hat{e}(X_i)[1-\hat{e}(X_j)]}, \\
\tilde{B}_4^{\text{AIPW}} &= \sum_{i=1}^n \sum_{j \neq i}^n \frac{Z_i(1-Z_j)}{\hat{e}(X_i)[1-\hat{e}(X_j)]}, \\
\frac{\partial r^{\text{AIPW}}(\cdot, \cdot; \hat{\beta}, \alpha)}{\partial \alpha} &= \frac{\frac{1}{2} \left\{ \frac{Z_i(1-Z_j)[- \partial \mu(X_i, X_j; \alpha)/\partial \alpha]}{\hat{e}(X_i)(1-\hat{e}(X_j))} + \frac{Z_j(1-Z_i)[- \partial \mu(X_j, X_i; \alpha)/\partial \alpha]}{\hat{e}(X_j)(1-\hat{e}(X_i))} \right\}}{\frac{1}{n(n-1)} \sum_{i=1}^n \sum_{j \neq i}^n Z_i(1-Z_j)/[\hat{e}(X_i)(1-\hat{e}(X_j))]} \\
&\quad + \frac{1}{2} \left[\frac{\partial \mu(X_i, X_j; \alpha)}{\partial \alpha} + \frac{\partial \mu(X_j, X_i; \alpha)}{\partial \alpha} \right], \\
C_1^{\text{AIPW}} &= \frac{\sum_{i=1}^n \sum_{j \neq i}^n Z_i(1-Z_j)[- \tilde{\mu}(X_i, X_j; \hat{\alpha})]/[\hat{e}(X_i)(1-\hat{e}(X_j))]}{\sum_{i=1}^n \sum_{j \neq i}^n Z_i(1-Z_j)/[\hat{e}(X_i)(1-\hat{e}(X_j))]} \\
&\quad + \frac{1}{n(n-1)} \sum_{i=1}^n \sum_{j \neq i}^n \tilde{\mu}(X_i, X_j; \hat{\alpha}),
\end{aligned}$$

where $\tilde{\mu}(X_i, X_j; \alpha)$ is defined in (A.19).

D3. expressions for B_1^{AOW} and C_1^{AOW}

When $\omega(X_i, X_j) = (1 - \hat{e}(X_i))\hat{e}(X_j)$, then we have $\hat{\tau}_1^{\text{AW}} = \hat{\tau}_1^{\text{AOW}}$, and relative detailed expressions are as follows:

$$\begin{aligned}
\hat{\tau}_1^{\text{AOW}} &= \binom{n}{2}^{-1} \sum_{i=1}^n \sum_{j>i}^n r^{\text{AOW}}(\cdot, \cdot; \hat{\beta}, \hat{\alpha}), \\
r^{\text{AOW}}(\cdot, \cdot; \hat{\beta}, \hat{\alpha}) &= \frac{\frac{1}{2}Z_i(1-Z_j)(1-\hat{e}(X_i))\hat{e}(X_j)[\mathbb{I}(Y_i > Y_j) - \hat{\mu}(X_i, X_j)]}{\frac{1}{n(n-1)} \sum_{i=1}^n \sum_{j \neq i}^n Z_i(1-Z_j)(1-\hat{e}(X_i))\hat{e}(X_j)} \\
&\quad + \frac{\frac{1}{2}Z_j(1-Z_i)(1-\hat{e}(X_j))\hat{e}(X_i)[\mathbb{I}(Y_j > Y_i) - \hat{\mu}(X_j, X_i)]}{\frac{1}{n(n-1)} \sum_{i=1}^n \sum_{j \neq i}^n Z_i(1-Z_j)(1-\hat{e}(X_i))\hat{e}(X_j)} \\
&\quad + \frac{\frac{1}{2}\hat{e}(X_i)(1-\hat{e}(X_i))\hat{e}(X_j)(1-\hat{e}(X_j))[\hat{\mu}(X_i, X_j) + \hat{\mu}(X_j, X_i)]}{\frac{1}{n(n-1)} \sum_{i=1}^n \sum_{j \neq i}^n \hat{e}(X_i)\hat{e}(X_j)(1-\hat{e}(X_i))(1-\hat{e}(X_j))}, \\
\hat{r}_1^{\text{AOW}}(\cdot; \hat{\beta}, \hat{\alpha}) &= (n-1)^{-1} \sum_{j \neq i}^n r^{\text{AOW}}(\cdot, \cdot; \hat{\beta}, \hat{\alpha}), \\
B_1^{\text{AOW}} &= \tilde{B}_1^{\text{AOW}} - \tilde{B}_3^{\text{AOW}} * \frac{\tilde{B}_2^{\text{AOW}}}{\tilde{B}_4^{\text{AOW}}} + \tilde{B}_5^{\text{AOW}} - \tilde{B}_7^{\text{AOW}} * \frac{\tilde{B}_6^{\text{AOW}}}{\tilde{B}_8^{\text{AOW}}}, \\
\tilde{B}_1^{\text{AOW}} &= \frac{\sum_{i=1}^n \sum_{j \neq i}^n Z_i(1-Z_j)(1-\hat{e}(X_i))\hat{e}(X_j)[(1-\hat{e}(X_j))X_j - \hat{e}(X_i)X_i]}{\sum_{i=1}^n \sum_{j \neq i}^n Z_i(1-Z_j)(1-\hat{e}(X_i))\hat{e}(X_j)} \times [\mathbb{I}(Y_i > Y_j) - \hat{\mu}(X_i, X_j)], \\
\tilde{B}_2^{\text{AOW}} &= \frac{\sum_{i=1}^n \sum_{j \neq i}^n Z_i(1-Z_j)(1-\hat{e}(X_i))\hat{e}(X_j)[\mathbb{I}(Y_i > Y_j) - \hat{\mu}(X_i, X_j)]}{\sum_{i=1}^n \sum_{j \neq i}^n Z_i(1-Z_j)(1-\hat{e}(X_i))\hat{e}(X_j)}, \\
\tilde{B}_3^{\text{AOW}} &= \sum_{i=1}^n \sum_{j \neq i}^n Z_i(1-Z_j)(1-\hat{e}(X_i))\hat{e}(X_j)[(1-\hat{e}(X_j))X_j - \hat{e}(X_i)X_i], \\
\tilde{B}_4^{\text{AOW}} &= \sum_{i=1}^n \sum_{j \neq i}^n Z_i(1-Z_j)(1-\hat{e}(X_i))\hat{e}(X_j), \\
\tilde{B}_5^{\text{AOW}} &= \frac{\sum_{i=1}^n \sum_{j \neq i}^n \hat{h}(X_i, X_j)[(1-2\hat{e}(X_i))X_i]\hat{\mu}(X_i, X_j)}{\sum_{i=1}^n \sum_{j \neq i}^n \hat{h}(X_i, X_j)}, \\
\tilde{B}_6^{\text{AOW}} &= \frac{\sum_{i=1}^n \sum_{j \neq i}^n \hat{h}(X_i, X_j)\hat{\mu}(X_i, X_j)}{\sum_{i=1}^n \sum_{j \neq i}^n \hat{h}(X_i, X_j)}, \\
\tilde{B}_7^{\text{AOW}} &= \sum_{i=1}^n \sum_{j \neq i}^n \hat{h}(X_i, X_j)[(1-2\hat{e}(X_i))X_i + (1-2\hat{e}(X_j))X_j], \\
\tilde{B}_8^{\text{AOW}} &= \sum_{i=1}^n \sum_{j \neq i}^n \hat{h}(X_i, X_j), \\
\frac{\partial r^{\text{AOW}}(\cdot, \cdot; \hat{\beta}, \alpha)}{\partial \alpha} &= \frac{\frac{1}{2}Z_i(1-Z_j)(1-\hat{e}(X_i))\hat{e}(X_j)[- \partial \mu(X_i, X_j; \alpha) / \partial \alpha]}{\frac{1}{n(n-1)} \sum_{i=1}^n \sum_{j \neq i}^n Z_i(1-Z_j)(1-\hat{e}(X_i))\hat{e}(X_j)} \\
&\quad + \frac{\frac{1}{2}Z_j(1-Z_i)(1-\hat{e}(X_j))\hat{e}(X_i)[- \partial \mu(X_j, X_i; \alpha) / \partial \alpha]}{\frac{1}{n(n-1)} \sum_{i=1}^n \sum_{j \neq i}^n Z_i(1-Z_j)(1-\hat{e}(X_i))\hat{e}(X_j)} \\
&\quad + \frac{\frac{1}{2}\hat{e}(X_i)(1-\hat{e}(X_i))\hat{e}(X_j)(1-\hat{e}(X_j))[\frac{\partial \mu(X_i, X_j; \alpha)}{\partial \alpha} + \frac{\partial \mu(X_j, X_i; \alpha)}{\partial \alpha}]}{\frac{1}{n(n-1)} \sum_{i=1}^n \sum_{j \neq i}^n \hat{e}(X_i)\hat{e}(X_j)(1-\hat{e}(X_i))(1-\hat{e}(X_j))},
\end{aligned}$$

$$C_1^{\text{AOW}} = \frac{\sum_{i=1}^n \sum_{j \neq i}^n Z_i(1 - Z_j)(1 - \hat{e}(X_i))\hat{e}(X_j)[- \tilde{\mu}(X_i, X_j; \hat{\alpha})]}{\sum_{i=1}^n \sum_{j \neq i}^n Z_i(1 - Z_j)(1 - \hat{e}(X_i))\hat{e}(X_j)} + \frac{\sum_{i=1}^n \sum_{j \neq i}^n \hat{h}(X_i, X_j)\tilde{\mu}(X_i, X_j; \hat{\alpha})}{\sum_{i=1}^n \sum_{j \neq i}^n \hat{e}(X_i)\hat{e}(X_j)(1 - \hat{e}(X_i))(1 - \hat{e}(X_j))},$$

where $\hat{h}(X_i, X_j) = \hat{e}(X_i)\hat{e}(X_j)(1 - \hat{e}(X_i))(1 - \hat{e}(X_j))$ and $\tilde{\mu}(X_i, X_j; \alpha)$ is defined in (A.19).

Web Appendix E: Addition Simulation design for covariates are strongly correlated

Similar to Section 5.1 of main text, we generate $p = 6$ baseline covariates, $\mathbf{X} = (X_1, X_2, \dots, X_6)^\top$, which come from multivariate normal distribution $N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, where mean vector $\boldsymbol{\mu} = (0, 0.2, 0.4, 0.6, 0.8, 1.0)^\top$ and variance-covariance matrix is

$$\boldsymbol{\Sigma} = \begin{pmatrix} 1.0 & 0.8 & 0.8 & 0.2 & 0.2 & 0.2 \\ 0.8 & 1.0 & 0.8 & 0.2 & 0.2 & 0.2 \\ 0.8 & 0.8 & 1.0 & 0.2 & 0.2 & 0.2 \\ 0.2 & 0.2 & 0.2 & 1.0 & 0.8 & 0.8 \\ 0.2 & 0.2 & 0.2 & 0.8 & 1.0 & 0.8 \\ 0.2 & 0.2 & 0.2 & 0.8 & 0.8 & 1.0 \end{pmatrix}.$$

Therefore, $X_1 - X_3$ are strongly correlated, $X_4 - X_6$ are also strongly correlated. Suppose participants are randomized into one of two groups, the treatment indicator is generated from $Z \sim \text{Bernoulli}(\pi)$ by fixing the randomization probability π . For illustrative purposes, we still assume the ordinal outcome has three levels with level 1 < level 2 < level 3, and that a higher level indicates a more favorable outcome. Based on baseline covariates $X_i = (X_{i1}, X_{i2}, \dots, X_{i6})^\top$, $i = 1, 2, \dots, n$, we generate the potential outcomes from the following model: for $z = 0, 1$,

$$Y_i(z) \sim \begin{cases} \text{logit}[\mathbb{P}(Y_i(z) \geq 2)] = \alpha_{10} + \sum_{j=1}^6 \gamma_j X_{ij} + \delta z, \\ \text{logit}[\mathbb{P}(Y_i(z) \geq 3)] = \alpha_{20} + \sum_{j=1}^6 \gamma_j X_{ij} + \delta z, \end{cases} \quad (\text{A.20})$$

where $\alpha_{10} = 1$, $\alpha_{20} = 0.05$, $\delta = 1$ and $\boldsymbol{\gamma} = (\gamma_1, \gamma_2, \gamma_3, \gamma_4, \gamma_5, \gamma_6)^\top = (1, -1, 1, -1, 1, -1)^\top \times (0.5 + 0.5z)$. Therefore, the parameter α in outcome $\mu(X_i, X_j; \alpha)$ can be characterized as $\alpha = (\alpha_{10}, \alpha_{20}, \boldsymbol{\gamma}^\top, \delta)^\top$.

After obtaining true values of WR θ and WD η by using Monte Carlo approximation based on the simulated potential outcome model (A.20), we compare several different estimators including the unadjusted estimator $\hat{\tau}^{\text{UNADJ}}$ (UNADJ), the IPW estimator $\hat{\tau}^{\text{IPW}}$ (IPW), the OW estimator $\hat{\tau}^{\text{OW}}$ (OW), the augmented IPW estimator $\hat{\tau}^{\text{AIPW}}$ (AIPW) and the augmented OW estimator $\hat{\tau}^{\text{AOW}}$ (AOW) for WR and WD. Similar to Section 5.2.1 and 5.2.2 in main text, we also evaluate the accuracy of the point estimator as well as the accuracy of the proposed variance estimators and the empirical coverage rates of the associated 95% confidence intervals, corresponding results are summarized in Table S2 and S3.

Reference

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Web Appendix F: An Additional Web Table

Table S1: Summary estimated win ratio (WR) and win difference (WD) by unadjusted estimators under two different variance approaches. AESE1 is calculated based on (16) of main text, AESE2 is calculated based on (5) of main text and AESE3 is calculated by the bias-reduced variance estimator suggested by Ozenne et al.(2021). CP1, CP2 and CP3 are corresponding coverage probability of the 95% confidence intervals. RB: relative bias; MCSD: Monte Carlo standard deviation; CP: coverage probability.

Model	Design	Sample size	Estimand	True	RB(%)	MCSD	AESE1	CP1	AESE2	CP2	AESE3	CP3
Quadratic	Balanced	25	WR	1.453	42.808	2.310	1.925	0.866	1.789	0.857	1.925	0.866
			WD	0.091	-13.187	0.194	0.188	0.926	0.178	0.907	0.187	0.922
		50	WR	1.453	20.716	1.276	1.078	0.893	1.049	0.891	1.078	0.893
			WD	0.091	-6.593	0.136	0.135	0.940	0.131	0.933	0.134	0.940
		100	WR	1.453	7.708	0.698	0.634	0.927	0.626	0.926	0.634	0.927
			WD	0.091	-4.396	0.097	0.096	0.952	0.095	0.952	0.096	0.952
		200	WR	1.453	4.749	0.437	0.428	0.932	0.426	0.931	0.428	0.932
			WD	0.091	2.198	0.068	0.068	0.949	0.067	0.948	0.068	0.949
		400	WR	1.453	2.546	0.298	0.294	0.946	0.293	0.946	0.294	0.946
			WD	0.091	2.198	0.048	0.048	0.941	0.048	0.941	0.048	0.941
	Unbalanced	25	WR	1.450	52.966	2.889	2.177	0.899	1.968	0.889	2.177	0.899
			WD	0.091	-2.198	0.215	0.210	0.913	0.191	0.887	0.201	0.898
		50	WR	1.450	26.000	1.282	1.128	0.919	1.092	0.914	1.128	0.919
			WD	0.091	10.989	0.151	0.148	0.947	0.143	0.929	0.146	0.935
		100	WR	1.450	7.586	0.705	0.660	0.922	0.651	0.919	0.660	0.922
			WD	0.091	-2.198	0.107	0.105	0.946	0.103	0.945	0.104	0.946
		200	WR	1.450	4.138	0.468	0.447	0.939	0.445	0.937	0.447	0.939
			WD	0.091	0.000	0.075	0.074	0.942	0.074	0.940	0.074	0.940
		400	WR	1.450	0.069	0.306	0.302	0.942	0.301	0.942	0.302	0.942
			WD	0.091	-4.396	0.053	0.053	0.951	0.052	0.950	0.052	0.951
Interaction	Balanced	25	WR	1.663	35.899	2.119	2.049	0.881	1.909	0.871	2.049	0.881
			WD	0.126	-9.524	0.188	0.190	0.929	0.179	0.905	0.189	0.926
		50	WR	1.663	18.882	1.322	1.199	0.908	1.165	0.904	1.199	0.908
			WD	0.126	-5.556	0.134	0.135	0.945	0.132	0.941	0.135	0.944
		100	WR	1.663	8.900	0.802	0.737	0.920	0.728	0.920	0.737	0.920
			WD	0.126	-2.381	0.097	0.096	0.937	0.095	0.933	0.096	0.937
		200	WR	1.663	4.811	0.517	0.492	0.933	0.489	0.932	0.492	0.933
			WD	0.126	0.794	0.068	0.068	0.946	0.067	0.943	0.068	0.945
		400	WR	1.663	2.345	0.341	0.336	0.948	0.335	0.948	0.336	0.948
			WD	0.126	0.794	0.048	0.048	0.949	0.048	0.949	0.048	0.949
	Unbalanced	25	WR	1.657	62.643	3.597	2.677	0.892	2.375	0.887	2.677	0.892
			WD	0.125	0.800	0.223	0.212	0.917	0.192	0.882	0.203	0.895
		50	WR	1.657	26.494	1.489	1.291	0.920	1.247	0.918	1.291	0.920
			WD	0.125	8.800	0.152	0.150	0.941	0.144	0.931	0.148	0.932
		100	WR	1.657	9.958	0.848	0.769	0.922	0.758	0.917	0.769	0.922
			WD	0.125	0.800	0.109	0.106	0.943	0.103	0.932	0.105	0.936
		200	WR	1.657	3.862	0.513	0.508	0.935	0.504	0.933	0.508	0.935
			WD	0.125	0.000	0.075	0.075	0.945	0.074	0.940	0.075	0.940
		400	WR	1.657	0.845	0.355	0.346	0.932	0.345	0.932	0.346	0.932
			WD	0.125	-1.600	0.054	0.053	0.940	0.053	0.936	0.053	0.938

Table S2: Relative bias (%) of estimated WR and WD from different estimators under balanced and unbalanced designs. UNADJ: the unadjusted estimator; IPW: inverse probability weighting; OW: overlap weighting; AIPW: augmented inverse probability weighting; AOW: augmented overlap weighting; AIPW-mis: augmented inverse probability weighting with a misspecified working outcome model; and AOW-mis: augmented overlap weighting with a misspecified working outcome model.

Design	n	Estimand	Truth	UNADJ	IPW	OW	AIPW	AOW	AIPW-mis	AOW-mis
Balanced	200	WR	1.546	2.17	1.20	1.29	1.09	1.19	1.20	1.27
		WD	0.131	-1.32	-2.14	-1.92	-2.25	-2.03	-2.07	-1.90
	300	WR	1.546	1.67	1.20	1.28	1.16	1.25	1.21	1.29
		WD	0.131	-0.03	-0.10	0.06	-0.09	0.08	-0.02	0.15
	400	WR	1.546	0.86	0.37	0.43	0.36	0.43	0.30	0.36
		WD	0.131	-0.80	-1.02	-0.88	-0.98	-0.83	-1.13	-0.99
Unbalanced	200	WR	1.538	3.39	2.25	2.33	2.27	2.35	2.32	2.35
		WD	0.129	1.10	-0.39	-0.25	-0.19	-0.04	-0.14	-0.15
	300	WR	1.538	2.63	2.41	2.57	2.25	2.47	2.31	2.51
		WD	0.129	1.98	2.16	2.46	1.94	2.35	2.01	2.39
	400	WR	1.538	1.08	1.09	1.30	1.08	1.18	1.15	1.28
		WD	0.129	-0.23	0.38	0.71	0.43	0.51	0.56	0.70

Table S3: The ratio between the average estimated variance over the true, empirical Monte Carlo variance ($\{\text{Est Var}\}/\{\text{MC Var}\}$), as well as the associated 95% confidence interval coverage rate of different methods. UNADJ: the unadjusted estimator; IPW: inverse probability weighting; OW: overlap weighting; AIPW: augmented inverse probability weighting; AOW: augmented overlap weighting; AIPW-mis: augmented inverse probability weighting with a misspecified working outcome model; and AOW-mis: augmented overlap weighting with a misspecified working outcome model.

n	$\{\text{Est Var}\}/\{\text{MC Var}\}$							95% confidence interval coverage						
	UNADJ	IPW	OW	AIPW	AOW	AIPW-mis	AOW-mis	UNADJ	IPW	OW	AIPW	AOW	AIPW-mis	AOW-mis
WR under balanced design														
200	0.985	0.955	0.963	0.942	0.954	0.947	0.958	0.950	0.935	0.938	0.938	0.937	0.934	0.937
300	1.006	0.987	0.985	0.983	0.986	0.985	0.986	0.940	0.941	0.942	0.936	0.938	0.941	0.942
400	0.971	0.990	0.992	0.985	0.988	0.994	0.996	0.935	0.933	0.933	0.934	0.934	0.932	0.932
WD under balanced design														
200	1.021	0.966	0.972	0.960	0.975	0.959	0.973	0.944	0.946	0.950	0.946	0.950	0.948	0.951
300	1.038	1.003	1.002	0.995	1.001	1.000	1.005	0.955	0.948	0.948	0.948	0.950	0.947	0.949
400	0.991	0.986	0.986	0.978	0.981	0.986	0.990	0.945	0.949	0.946	0.942	0.942	0.950	0.948
WR under unbalanced design														
200	1.025	0.977	0.993	0.978	1.051	0.976	1.032	0.951	0.933	0.942	0.935	0.951	0.936	0.947
300	1.022	0.957	0.978	0.964	1.033	0.961	1.013	0.955	0.947	0.951	0.948	0.954	0.950	0.951
400	1.011	0.963	0.953	0.958	1.004	0.962	0.992	0.946	0.947	0.945	0.946	0.947	0.945	0.949
WD under unbalanced design														
200	1.062	1.016	1.031	0.998	1.088	0.991	1.066	0.957	0.952	0.958	0.950	0.960	0.944	0.959
300	1.061	0.994	1.024	0.981	1.075	0.980	1.059	0.959	0.948	0.958	0.945	0.965	0.945	0.960
400	1.048	1.008	0.994	0.983	1.042	0.987	1.029	0.956	0.949	0.948	0.944	0.949	0.942	0.950

Table S4: Empirical Type I error rate of estimated WR and WD from different estimators under balanced and unbalanced designs. UNADJ: the unadjusted estimator; IPW: inverse probability weighting; OW: overlap weighting; AIPW: augmented inverse probability weighting; AOW: augmented overlap weighting; AIPW-mis: augmented inverse probability weighting with a misspecified working outcome model; and AOW-mis: augmented overlap weighting with a misspecified working outcome model.

Design	n	Estimand	UNADJ	IPW	OW	AIPW	AOW	AIPW-mis	AOW-mis
Balanced	200	WR	0.053	0.058	0.058	0.054	0.058	0.057	0.056
		WD	0.055	0.059	0.060	0.057	0.058	0.058	0.056
	300	WR	0.041	0.053	0.052	0.054	0.055	0.050	0.049
		WD	0.042	0.053	0.054	0.055	0.055	0.051	0.051
	400	WR	0.051	0.048	0.048	0.048	0.048	0.053	0.054
		WD	0.052	0.048	0.048	0.049	0.049	0.055	0.054
Unbalanced	200	WR	0.047	0.046	0.054	0.057	0.048	0.049	0.049
		WD	0.050	0.048	0.053	0.062	0.050	0.053	0.050
	300	WR	0.049	0.060	0.059	0.059	0.052	0.061	0.047
		WD	0.049	0.062	0.060	0.063	0.050	0.062	0.050
	400	WR	0.055	0.056	0.055	0.059	0.045	0.053	0.046
		WD	0.057	0.057	0.056	0.062	0.047	0.056	0.047